Samir Novak

Senior ML Systems Engineer | Prague / remote EU
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TECH / PLATFORMS

Python, C++

PyTorch; JAX (research prototypes)

Docker / Kubernetes

Terraform, GitHub Actions

GCP (GKE, BigQuery)

Kafka, Apache Spark

PostgreSQL, Airflow

LANGUAGES

Czech — native

English — fluent (C1)

German — intermediate (B1)

EDUCATION

PhD, Computational Physics — Charles University, 2015–2019

MSc, Applied Mathematics — Czech Technical University, 2013–2015

EXPERIENCE

Aster Robotics — Staff ML Platform Engineer

08/2023–present | Prague / hybrid

• Own the training platform used by perception teams: Python services on Kubernetes, image builds with Docker, infrastructure changes through Terraform and GitHub Actions.

• Moved GPU training jobs from ad-hoc VMs to GKE; introduced experiment metadata in PostgreSQL and scheduled data-preparation workflows with Airflow.

• Partner with researchers using PyTorch; supported one JAX-based prototype without making JAX a production dependency.

Mercury Forecasting — ML Engineer / Data Engineer

Jan 2020–Jul 2023

• Built demand-forecasting models in PyTorch and maintained Python/C++ feature-generation components for low-latency scoring.

• Processed event streams with Kafka and Spark; training datasets landed in BigQuery, while operational state lived in PostgreSQL.

• Containerized internal services and helped the team adopt Kubernetes for scheduled inference workloads.

Charles University — Doctoral Researcher

2015–2019

Developed simulation and optimization code in Python and C++; later used early PyTorch versions for differentiable surrogate models. Research was carried out as part of the PhD in Computational Physics.

SELECTED NOTES

• Regularly review architecture proposals and mentor two engineers, but remain an individual contributor.

• Comfortable across model-development and platform layers; strongest production experience is in Python, Kubernetes and GCP.